

Sakhinala Sanjay Bhargav

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Summary — AI/ML Engineer specializing in reinforcement learning and production ML systems. Reduced electromagnetic simulation costs by 65% through novel RL-surrogate model architecture. Published researcher in 6G antenna optimization with hands-on expertise in NLP, computer vision, and end-to-end model deployment.

Skills

- **ML & AI:** Reinforcement Learning (SAC, PPO), Deep Learning, Computer Vision, Transformers, RAG, LLMs
- **Frameworks:** PyTorch, TensorFlow, HuggingFace, Stable-Baselines3, Scikit-Learn, OpenCV, LightGBM
- **NLP:** Multilingual NLP, Transformer Fine-Tuning, Semantic Search, OCR, Cross-Encoder Reranking, LangChain
- **Engineering:** Python, Docker, FastAPI, Linux, SQL, Git, Google Cloud Platform
- **Data & Tools:** Pandas, NumPy, FAISS, Optuna, Redis, Matplotlib, Seaborn

Projects

Reinforcement Learning for Antenna Design Optimization

[Blog Post](#)

- Reduced CST electromagnetic simulation calls by 65% using Soft Actor-Critic RL agent with blended surrogate models (LightGBM + MLP + RidgeCV), achieving **f0 MAE ~0.02 GHz** and **S11 MAE ~0.72 dB**.
- Built custom Gymnasium environment with uncertainty-aware reward shaping for 28 GHz antenna optimization, achieving target impedance matching in 40% fewer training episodes than baseline approaches.
- Developed hybrid surrogate model architecture with iterative CST validation to ensure fabrication feasibility for 6G applications.
- **Tech:** PyTorch, CUDA, Stable-Baselines3, LightGBM, Optuna, CST Studio API.

IndicRAG — Multilingual Document QA System

[GitHub](#) — [Blog Post](#)

- Built RAG pipeline supporting 12 Indian languages using LaBSE embeddings and cross-encoder reranking, achieving **78.3% F1 Score** on XQuAD (Hindi/Telugu) benchmarks.
- Reduced model size by 4x using INT8 quantization while maintaining accuracy, enabling efficient deployment.
- Integrated OCR preprocessing pipeline (Tesseract + LayoutParser) for scanned documents and deployed scalable FastAPI service with FAISS vector retrieval.
- **Tech:** HuggingFace Transformers, FAISS, PyTesseract, FastAPI, Redis, Docker.

CipherChat — Secure Encrypted Messenger

[GitHub: CipherChat](#)

- Built multi-threaded CLI messaging system using AES-256 and RSA-2048 hybrid cryptography with Diffie-Hellman key exchange.
- Implemented encrypted TCP sockets with SHA-256 integrity verification for concurrent multi-user messaging.
- **Tech:** Python, Socket Programming, Cryptography, TCP/IP, Multi-threading.

EuroSAT Land Use Classification

[GitHub: landuse-cnn-eurosat](#)

- Developed CNN achieving 89% test accuracy across 10 land-use classes on satellite imagery using transfer learning.
- Integrated Grad-CAM visualization for model interpretability and feature attribution analysis.
- **Tech:** PyTorch, TorchVision, OpenCV, Grad-CAM.

ResolverLab — DNS Resolver Benchmarking & Filtering Analysis

[GitHub](#)

- Designed high-concurrency DNS benchmarking framework to evaluate latency, reliability, and block accuracy across public resolvers and filtered DNS services.
- Implemented multi-protocol support (UDP, TCP, DNS-over-HTTPS, DNS-over-TLS) with asynchronous execution to scale testing across thousands of domains.
- Built automated blacklist ingestion pipeline and generated comparative HTML reports analyzing resolver behavior, error rates, and category-level filtering performance.
- **Tech:** Python, asyncio, DNS Protocols (UDP/TCP/DoH/DoT), aiohttp, HTML Reporting.

Industry Training & Externships

SkillHive Connect — Generative AI Training Program <i>Generative AI Training Participant</i> - Completed hands-on training in Generative AI, ML/DL, and NLP using Python. - Built and fine-tuned LLM-based applications using Transformers, LangChain, RAG, and LLaMA-derived models. - Developed applied GenAI projects including chatbots, semantic search, transcript summarization, and a voice-based medical assistant.	Aug 2025 – Dec 2025
Google for Developers — AI/ML Training Program <i>AI/ML Training Participant</i> - Completed hands-on training in computer vision using Python and OpenCV for mobile application development. - Built end-to-end ML deployment pipelines using Vertex AI APIs, from model training to production inference. - Developed real-time inference workflows integrating ML models into application backends.	Jan 2025 – Mar 2025
Palo Alto Networks — Cybersecurity Training Program <i>Cybersecurity Training Participant</i> - Completed training in firewall policy optimization and network access control configuration. - Analyzed threat logs and SIEM workflows for anomaly detection using log-correlation techniques. - Gained practical knowledge of secure network architecture, intrusion detection systems, and endpoint protection.	Oct 2024 – Dec 2024
Altair — Data Science Training Program <i>Data Science Training Participant</i> - Built automated ML pipelines in RapidMiner for data preprocessing, feature engineering, and model evaluation. - Benchmarked supervised learning algorithms (SVM, Random Forest) for classification tasks and performance analysis. - Performed exploratory data analysis and statistical modeling to derive actionable insights.	Apr 2024 – Jun 2024

Education

B.Tech in Electronics and Communication Engineering <i>Bapatla Engineering College, Bapatla, Andhra Pradesh</i>	May 2025
Diploma in Electronics and Communication Engineering <i>Aditya College of Engineering, Surampalem, Andhra Pradesh</i>	Oct 2021

Certifications

- Google Data Analytics Professional Certificate** — Completed Mar 2024
SQL, R Programming, Data Visualization (Tableau), Statistical Analysis
- Google Cybersecurity Professional Certificate** — Completed May 2024
Network Security, Python for Security, Linux Essentials, Cloud Security
- Google Prompting Essentials** — Completed May 2024
Prompt engineering techniques for GenAI workflows
- AI For Everyone** — DeepLearning.AI — Completed Feb 2025
AI strategy, ML fundamentals, business applications
- Google Cloud Skill Badges (10+)**
TensorFlow, Image Classification, Object Detection, ML Pipelines, Cloud Innovator Program
- NPTEL Certifications**
Cloud Computing; Computer Networks & Internet Protocols

Publications

- Bandwidth Enhancement of Slotted Hexagonal Patch Antenna for 6G Ultra-Fast Data Transfer and Brain-Computer Interface**
Presented at ICMOCE 2025, IIT Bhubaneswar
DOI: 10.1109/ICMOCE64100.2025.11076991 — Published July 2025
- Bandwidth Optimization of Slotted Circular Patch Antenna for 6G Ultra-Fast Data Transfer and Brain-Computer Interface Application**
Presented at 16th ICCCNT 2025, IIT Indore
Proceedings submitted to IEEE Xplore — Publication Pending
- Uncertainty-Aware Reinforcement Learning System with Blended Surrogate Models for Electromagnetic Structure Optimization**
Under Review

Patents

- Design of Hexagonal Patch Antenna at 28 GHz**
Inventor: Sakhinala Sanjay Bhargav
Application No.: 202541014595 A — Published Mar 2025 (India)